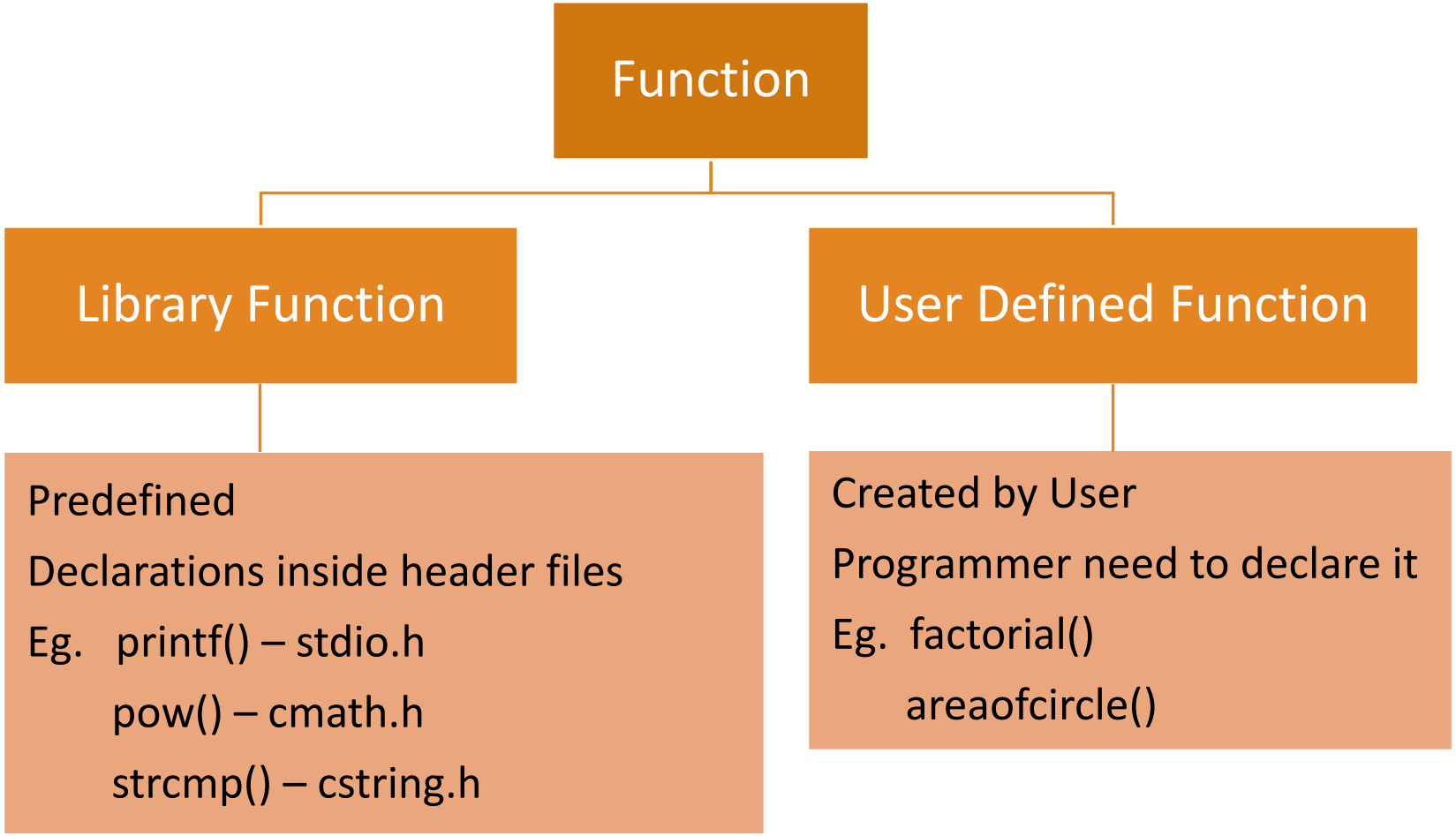
Chapter - 4

FUNCTIONS IN C++

C++ Function

A **function** is a group of statements that together perform a task.

Functions are made for code reusability and for saving time and space.



C++ Function – (Cont…)

There are three elements of user defined function

```
void func1();
void main()
{
    ....
    func1();
}
void func1()
{
    ....
    ....
}
```

Function Declaration

Function call

Function body

Function definition

Simple Function – (Cont…)

- **Function Declaration**

Syntax:

```
return-type function-name (arg-1, arg 2, …);
```

Example: `int addition(int , int );`

- **Function Definition**

Syntax:

```
return-type function-name (arg-1, arg 2, …)
{
     ... Function body
}
```

Example:

```
int addition(int x, int y)
{
  return x+y;
}
```

Categories of Function

(1) Function **with arguments** and **returns** value

Function arguments/ parameters

Return type

```
int func1(int , int ); \\declaration
void main()
{
   ....
   int z = func1(5,6); \\function call
}
int func1(int a, int b) \\definition
{
   ....
   return a+b;
}
```

Function **func1** returns integer value to variable z

returns a+b to calling function

Categories of Function (Cont…)

(2) Function **with arguments** but **no return** value

```
void func1(int , int ); \\function declaration
void main()
{
   ....
   func1(5,6); \\function call
}
void func1(int a, int b) \\function definition
{
   ....
   ....
}
```

Categories of Function (Cont..)

(3) Function with **no argument** but **returns** value

```
int func1();
void main()
{
   ....
   int z = func1();
}
int func1()
{
   ....
   return 99;
}
```

Categories of Function (Cont..)

(3) Function with **no argument** but **returns** value

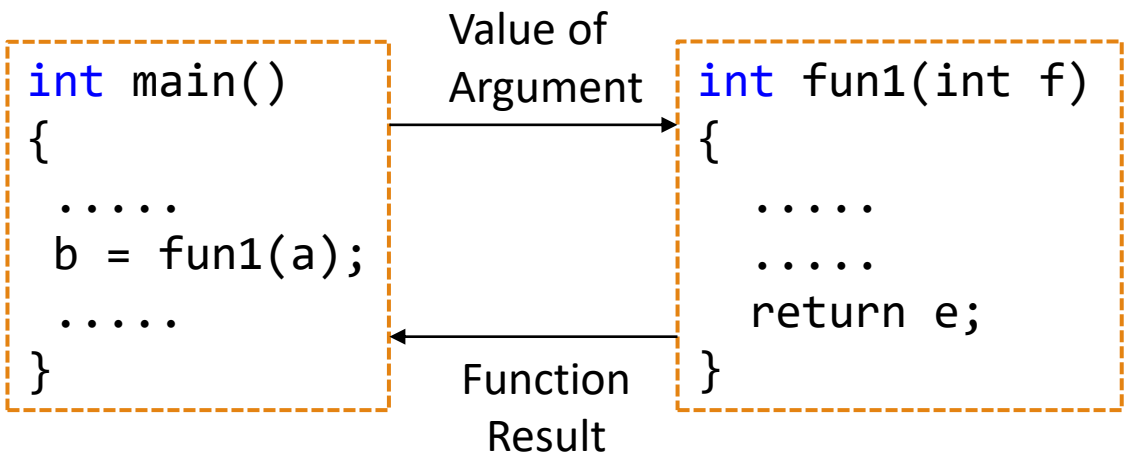
```
int func1();
void main()
{
   ....
   int z = func1();
}
int func1()
{
   ....
   return 99;
}
```

Program: Categories of function

Write C++ programs to demonstrate various categories of function, Create function **addition** for all categories.

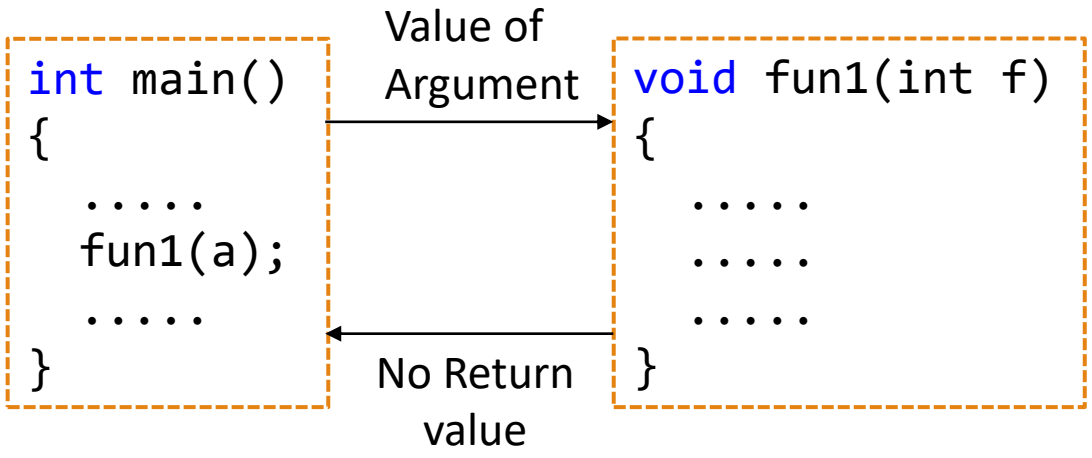
Function with argument and returns value

```
#include <iostream>
using namespace std;

int add(int, int);

int main(){
  int a=5,b=6,ans;
  ans = add(a,b);
  cout<<"Addition is="<<ans;
  return 0;
}
int add(int x,int y)
{
  return x+y;
}
```



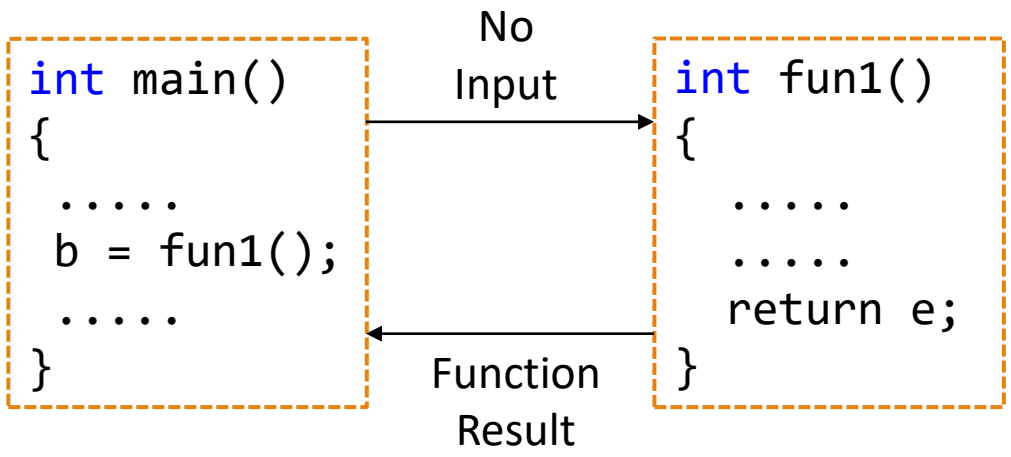
Function with arguments but no return value

```
#include <iostream>
using namespace std;

void add(int, int);
int main()
{
  int a=5,b=6;
  add(a,b);
  return 0;
}
void add(int x,int y)
{
 cout<<"Addition is="<<x+y;
}
```



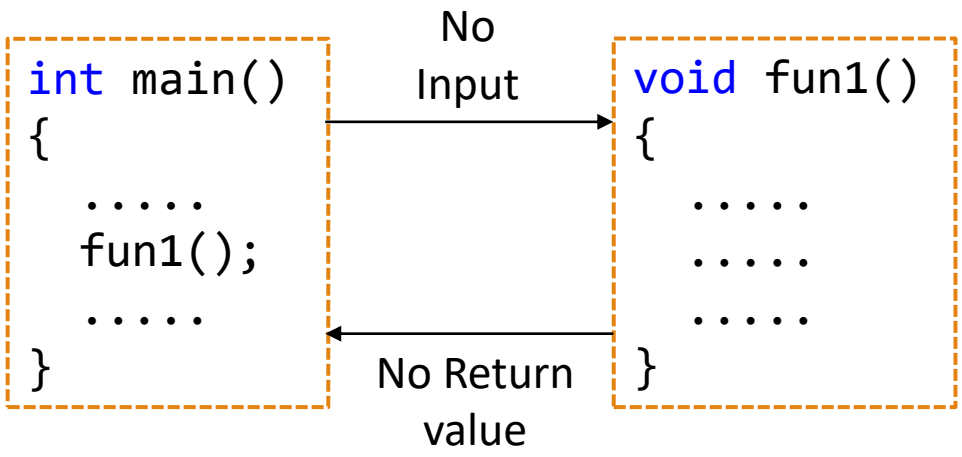
Function with no argument but returns value

```
int add();

int main()
{
  int ans;
  ans = add();
  cout<<"Addition is="<<ans;
  return 0;
}
int add()
{
  int a=5,b=6;
  return a+b;
}
```



Function with no argument and no return value

```
void add();

int main()
{
  add();
  return 0;
}
void add()
{
  int a=5,b=6;
  cout<<"Addition is="<<a+b;
}
```



Categories of Functions Summary

(1) Function with argument and returns value

```
int main()
{
 .....
 b = fun1(a);
 .....
}
```

Value of Argument →

```
int fun1(int f)
{
  .....
  .....
  return e;
}
```

← Function Result

(2) Function with argument and but no return value

```
int main()
{
  .....
  fun1(a);
  .....
}
```

Value of Argument →

```
void fun1(int f)
{
  .....
  .....
  .....
}
```

← No Return value

Categories of Functions Summary

(3) Function with no argument and returns value

```
int main()
{
 .....
 b = fun1();
 .....
}
```

No Input →

```
int fun1()
{
   .....
   .....
   return e;
}
```

← Function Result

(4) Function with no argument and but no return value

```
int main()
{
   .....
   fun1();
   .....
}
```

No Input →

```
void fun1()
{
   .....
   .....
   .....
}
```

← No Return value

Call by Reference

pass by reference

cup =

fillCup( )

pass by value

cup =

fillCup( )

Call by reference

The **call by reference** method of passing arguments to a function copies the reference of an argument into the formal parameter.

Inside the function body, the reference is used to access the actual argument used in the call.

Actual Parameters

```
int main(){
   add(a,b);
}

void add(int x,int y){
   cout << x+y;
}
```

Formal Parameters

Note:

- *Actual parameters* are parameters as they appear in function calls.
- *Formal parameters* are parameters as they appear in function declarations / definition.

Program: Swap using pointer, reference

Write a C++ program that to swap two values using function

1. With pass by pointer
2. With pass by reference

Program: Solution

```
void swapptr(int *x, int *y)
{
  int z = *x;
  *x=*y;
  *y=z;
}
void swapref(int &x, int &y)
{
  int z = x;
  x = y;
  y = z;
}
```

- Pointers as arguments
- References as arguments

```
int main()
{
  ...
  swapptr(&a,&b);
  swapref(a,b);
  ...
}
```

Program: Solution

```
void swapptr(int *, int *);
void swapref(int &, int &);
int main()
{
  int a = 45;
  int b = 35;
  cout<<"Before Swap\n";
  cout<<"a="<<a<<" b="<<b<<"\n";

  swapptr(&a,&b);
  cout<<"After Swap with pass by pointer\n";
  cout<<"a="<<a<<" b="<<b<<"\n";

  swapref(a,b);
  cout<<"After Swap with pass by reference\n";
  cout<<"a="<<a<<" b="<<b<<"\n";
}
```

Program: Solution (Cont…)

```
void swapptr(int *x, int *y)
{
  int z = *x;
  *x=*y;
  *y=z;
}
void swapref(int &x, int &y)
{
  int z = x;
  x = y;
  y = z;
}
```

OUTPUT

```
Before Swap
a=45 b=35
After Swap with pass by pointer
a=35 b=45
After Swap with pass by reference
a=45 b=35
```

Program: Return by Reference

Write a C++ program to **return reference** of maximum of two numbers from function max.

Program: Solution

```
int& max(int &, int &);
int main()
{
    int a=5,b=6,ans;
    ans = max(a,b);
    cout<<"Maximum="<<ans;
}
int& max(int &x,int &y)
{
    if (x>y)
      return x;
    else
      return y;
}
```

- Function declaration returning reference

Program: Returning Reference

```
int x;
int& setdata();
int main()
{
    setdata() = 56;
    cout<<"Value="<<x;
    return 0;
}
int& setdata()
{
    return x;
}
```

- setx() is declared with a reference type,

  `int&`

  as the return type:
- `int& setx();`

  This function contains

  `return x;`
- You can put a call to this function on the left side of the equal sign:

  `setx() = 92;`
- The result is that the variable returned by the function is assigned the value on the right side of the equal sign.

Inline Functions

Every time a function is called it takes a lot of extra time to execute series of instructions such as

1. Jumping to the function
2. Saving registers
3. Pushing arguments into stack
4. Returning to the calling function

If a function body is small then overhead time is more than actual code execution time so it becomes more time consuming.

Preprocessor macros is a solution to the problem of small functions in C.

In C++, **inline function** is used to reduce the function call overhead.

Inline Functions (Cont…)

Syntax:

```
inline return-type function-name(parameters)
{
    // function code
}
```

- Add **inline** word before the function definition to convert simple function to inline function.

Example:

```
inline int Max(int x, int y)
{
    if (x>y)
      return x;
    else
      return y;
}
```

Program: Inline function

Write a C++ program to create inline function that returns cube of given number (i.e **`n=3, cube=(n*n*n)=27`**).

Program: Solution

```
#include <iostream>
using namespace std;

inline int cube(int s)
{
    return s*s*s;
}
int main()
{
    cout << "The cube of 3 is: " << cube(3);
    return 0;
}
```

- Calls inline function cube with argument 3

Critical situations Inline Functions

Some of the situations inline expansion may not work

1) If a **loop**, a **switch** or a **goto** exists in function body.
2) If function is not returning any value.
3) If function contains **static variables**.
4) If function is **recursive**.

Function Overloading

- Suppose we want to make functions that add 2 values, add 3 values , add 4 values

In C

```
int sum(int a, int b);
int sum(int a, int b, int c);
int sum(int a, int b, int c, int d);
```

Function with same name in a program **is not allowed** in C language

In C++

```
int sum(int a, int b);
int sum(int a, int b, int c);
int sum(int a, int b, int c, int d);
```

Function with same name in a program **is allowed** in C++ language

Function overloading – Cont…

- C++ provides **function overloading** which allows to use multiple functions **sharing the same name** .
- Function overloading is also known as **Function Polymorphism** in OOP.
- It is the practice of declaring the same function with **different signatures**.

- However, the two functions with the same name must differ in at least one of the following,
  a) The **number** of arguments
  b) The **data type** of arguments
  c) The **order** of appearance of arguments

Arguments make the function unique

Function overloading does not depends on return type.

Function Overloading

`int sum(int a, int b);`	Valid
`float sum(int a, int b);`	Invalid
`int sum(int a, int c);`	Invalid
`int sum(int a, float b);`	Valid
`int sum(float b, int a);`	Valid
`float sum(float a, float b);`	Valid
`int sum(int a, int b, int c);`	Valid
`int sum(int a, float b, int c);`	Valid

Program: Function overloading

Write a C++ program to demonstrate function overloading. Create function **display()** with different arguments but same name

Program: Solution (Cont…)

```
void display(int var)
{
   cout << "Integer number: " << var << endl;
}
void display(float var)
{
   cout << "Float number: " << var << endl;
}
void display(int var1, float var2) {
   cout << "Integer number: " << var1;
   cout << " and float number:" << var2;
}
```

Program: Solution

```
int main()
{
   int a = 5; float b = 5.5;
   display(a);
   display(b);
   display(a, b);
   return 0;
}
```

Program: Function overloading

Write a C++ program to demonstrate function overloading. Create function **`area()`** that calculates area of circle, triangle and box.

Program #7 Solution

```
float area(int r)
{
  return 3.14*r*r;
}
float area(int h, int b)
{
  return 0.5*h*b;
}
float area(int l, int w, int h)
{
  return l*w*h;
}
int main(){
   cout<<"area of circle="<<area(5);
   cout<<“\n area of triangle="<<area(4,9);
   cout<<“\n area of box="<<area(5,8,2);
   return 0;
}
```

Default Function Argument

```
int cubevolume(int l=5, int w=6, int h=7)
{
   return l*w*h;
}
```

```
int main()
{
cubevolume();
cubevolume(9);
cubevolume(15,12);
cubevolume(3,4,7);
}
```

Here, there can be four types of function calls possible

If argument is not specified then, compiler looks at function to see how many arguments a function uses and alert program to use default values

Default Argument Example

```
int volume(int l=5,int w=6, int h=7)
{
   return l*w*h;
}
int main() {
   cout<<"volume="<<volume()<<endl;
   cout<<"volume="<<volume(9)<<endl;
   cout<<"volume="<<volume(15,2)<<endl;
   cout<<"volume="<<volume(3,4,7)<<endl;
   return 0;
}
```

- Function call passing all arguments.
- Explicitly value **3,4,7** passed to **l,w,h** respectively.
- Default value **7** considered for **h** respectively.

Default Arguments

while invoking a function If the argument/s are not passed then, the default values are used.

We must add default arguments from right to left.

We cannot provide a default value to a particular argument in the middle of an argument list.

Default arguments are useful in situations where some arguments always have the same value.

```
int cubevolume( int l    , int w = 2, int h    )
{
   return l*w*h;
}
```

Default Arguments (Cont…)

Legal and illegal default arguments

```
void f(int a, int b, int c=0);
void f(int a, int b=0, int c=0);
void f(int a=0, int b, int c=0);
void f(int a=0, int b, int c);
void f(int a=0, int b=0, int c=0);
```

Valid

Valid

Invalid

Invalid

Valid

Common Mistakes

(1)
```
void add(int a, int b = 3, int c, int d = 4);
```

- You cannot miss a default argument in between two arguments.
- In this case, **c** should also be assigned a default value.

(2)
```
void add(int a, int b = 3, int c, int d);
```

- If you want a single default argument, make sure the argument is the last one.

Program: Default Arguments

Write a C++ program to create function **`sum()`**, that performs addition of 3 integers also demonstrate Default Arguments concept.

Program: Default Arguments

```
#include <iostream>
using namespace std;
int sum(int x, int y=10, int z=20)
{
   return (x+y+z);
}
int main()
{
   cout << "Sum is : " << sum(5) << endl;
   cout << "Sum is : " << sum(5,15) << endl;
   cout << "Sum is : " << sum(5,15,25) << endl;
   return 0;
}
```

Friend Function

A friend function is not a member of a class but has access to private and protected members of the class.

Declared using the keyword friend inside the class.

It can be declared:

- Inside the class using friend
- Defined outside the class without scope resolution

Typically used for:

- Operator overloading
- Non-member functions needing access to class internals

Example

```
class ClassName {
   friend return_type function_name(arguments);
};
```

Key Points About Friend Function

It is not called with object syntax: object.function() doesn’t apply.

It does not support inheritance.

Used only when tight coupling is necessary.

Breaks encapsulation – use sparingly.

Virtual Function

A virtual function is a member function declared using the keyword virtual.

Allows a derived class to override a base class function.

Ensures the correct function is called during runtime.

Why Virtual Function?

- Used in **runtime polymorphism** (late binding).
- Essential when using **base class pointers** to refer to **derived objects**.

Example

```
class Base {
public:
    virtual void show() {
        cout << "Base class show()" << endl;
    }
};
```

const Arguments

In C++, an argument to a function can be declared as **const** as shown below:

```
int strlen(const char *p);
int length(const string &s);
```

The qualifier **const** tells the compiler that the function should not modify the argument.

Math Library functions

Table 4.1 *Commonly used math library functions*

Function	*Purposes*
ceil(x)	Rounds x to the smallest integer not less than x ceil(8.1) = 9.0 and ceil(-8.8) = –8.0
cos(x)	Trigonometric cosine of x (x in radians)
exp(x)	Exponential function e^x.
fabs(x)	Absolute value of x. If x>0 then abs(x) is x If x=0 then abs(x) is 0.0 If x<0 then abs(x) is –x
floor(x)	Rounds x to the largest integer not greater than x floor(8.2) = 8.0 and floor(–8.8 = –9.0
log(x)	Natural logarithm of x(base e)
log10(x)	Logarithm of x(base 10)
pow(x,y)	x raised to power y(x^y)
sin(x)	Trigonometric sine of x (x in radians)
sqrt(x)	Square root of x
tan(x)	Trigonometric tangent of x (x in radians)

note

The argument variables **x** and **y** are of type **double** and all the functions return the data type **double**.

To use the math library functions, we must include the header file **math.h** in conventional C++ and **cmath** in ANSI C++.

Open Parameter list / variable argument list

Refer to the program

Thank You